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A single day of constant light (L/L) provides immunity to behavioral despair in female rats maintained on an L/D cycle

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Abstract

The present experiment investigated the potentially ameliorative effect of exposure to light in the dark phase of an 12:12 h daily lighting schedule (12L/12D cycle) on behavioral despair, an animal model of depression based on two forced swim tests separated by 24 h. Experimental groups of female Wistar rats were maintained on the 12L/12D cycle except for a single exposure to 12 h of light treatment in the dark phase of the 12L/12D cycle. Control animals were treated similarly except for light treatment. Animals then underwent one of two sets of behavioral tests starting on either the day light (or control) treatment ended (No Delay groups) or 24 h thereafter (Delay groups). The treatment for subgroups of light-treated and control animals tested with or without delay consisted of either two forced swim tests separated by 24 h or testing in the open field and elevated plus maze. Results indicated that a single exposure to a 12-h light treatment has protective effect on behavioral despair in groups tested with or without delay as measured by shorter duration of immobility in the second swim test compared to the controls. Light-treated and control animals behaved similarly in the open field and elevated plus-maze tests.
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1. Introduction

While advances have been made in phototherapy, treatment of some forms of depression such as Seasonal Affective Disorder with exposure to bright light, the mechanism by which light can ameliorate depressive symptoms is not fully understood (Avery, 1998; Goodwin et al., 1982; Lewy et al., 1982, 1987; Rosenthal et al., 1984; Rosenthal and Wehr, 1992; Wehr and Wirz-Justice, 1982). Animal models of depression may provide the means to elucidate the ameliorative effect of light action in depression. There is, however, surprisingly little research on the effect of light on affective behavior in animals. A study by Molina-Hernandez and Tellez-Alcantara (2000) has shown that a prolonged treatment with a long photoperiod (14L/10D light/dark cycle) has an antidepressant effect as measured by reduced immobility in a forced swim test comparable to treatment with antidepressants.

In the light of the Molina-Hernandez and Tellez-Alcantara (2000) study showing ameliorative effect of light on immobility in forced swimming, female rats were exposed to a single treatment of light in the dark phase of an L/D cycle with the expectation that such photic stimulation will have a protective effect on behavioral despair, an animal model of depression. The model, originally developed by Porsolt et al. (1977, 1978) and reviewed by Willner (1990), is based on two forced swim tests separated by 24 h. Rats and mice initially display struggling behavior that eventually gives way to immobility, that occurs earlier, and for a longer duration in the second swim test than in the comparable period in the first test. Many antidepressants used to treat depression in humans decrease immobility in the second swim test compared to

Abbreviations: FST, forced swim test; HPA, hypothalamo-pituitary-adrenal axis; SAD, seasonal affective disorder; SCN, suprachiasmatic nucleus.

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controls (Armario et al., 1988; Connor et al., 2000; Dalvi and Lucki, 1999; Lucki, 1997; Porsolt et al., 1977, 1978; Reneric and Lucki, 1998).

An additional aim of the present study was to investigate the effect of light treatment on female rats in light of the fact that gender affects the incidence and severity of depression. There is a difference in both the frequency of occurrence and various manifestations of depression in females and males (Bracke, 1998; Dam et al., 1998; Lee and Chan, 1998; Nilssen et al., 1999). Comparative studies on animal models of depression also indicate sex differences (Barros and Ferigolo, 1998; Brotto et al., 2000; Kennett et al., 1986). On the other hand, studies utilizing animal models of depression, particularly the rat model of behavioral despair, have mostly ignored female rats possibly because of the potential confounding effect of the oestrus cycle in forced swimming. However, while an important issue, particularly with repeated exposure to forced swimming (Contreras et al., 1998), oestrus cycle does not seem to alter behavioral despair critically (Alonso et al, 1991). The present study therefore investigated the effect of an alteration in the photoperiod of female Wistar rats on the performance in forced swim tests. Additionally, the specificity of light treatment on behavioral despair was investigated by assessing the performance of independent groups of light-treated and control rats in the open field and elevated plus-maze tests.

2. Methods

2.1. Animals

A total of 56 female Wistar rats from our breeding colony, weighing 195–225 grams, were used in the experiments. Subjects were group housed in plastic cages. They were maintained on a 12 h light/12 h dark cycle (lights on at 0700 h) in a temperature controlled vivarium (22±2 °C) except for a single treatment in an isolated experimental chamber as described below. Food and water were available ad libitum.

2.2. Apparatus

Light treatment was applied in an insulated, ventilated chamber with a 25-tungsten lamp as source of illumination (approximately 50 cm above the top of the cage holding the subjects).

2.3. Experimental procedure

Animals lived in the vivarium in the home cages under the regular 12L/12D cycle except for the day of light or control treatment. For treatment, subjects were taken from their home cages 10 min before the beginning of the dark period in the L/D cycle and introduced into the insulated chamber in pairs, each pair housed in a plastic cage (two pairs in two cages were run on any given day). Animals had free access to food and water in the chamber. Animals in the light treatment groups were exposed to light in the chamber between 1900 and 0700 h when they would normally be in the dark phase of the L/D cycle. These animals therefore lived under a 24-h light regimen (L/L) for 1 day. Control animals were kept in the chamber but were not exposed to light in the dark phase; these animals lived their regular L/D cycle. Light was turned on at 0700 in the chamber for both groups, thus starting the light phase of the next L/D cycle. All animals were then removed from the chamber at 1500 h and were tested as described below either immediately afterwards (no delay condition) or at 1500 h the next day (24-h delay condition). Animals in both groups were returned to their home cages in the vivarium either after the tests specified below (no delay groups) or after a similar amount of time had passed (24-h delay groups). After the aforementioned treatment, all subjects lived under the regular 12L/12D cycle when they were returned to their home cages in the vivarium. Of the 56 subjects, 32 belonging to the experimental and control groups in both the delay and no delay conditions were administered two forced swim tests (n=8 per condition) separated by 24 h. The rest of the animals underwent two behavioral tests (open field and elevated plus-maze tests) in a counterbalanced fashion. There were eight animals in the light-treated groups tested with or without delay whereas the control groups consisted of four animals for each delay condition: these were then combined as the control group for statistical testing. The two tests were separated by 2 h. All tests were recorded on videotape.

2.4. Forced swim tests

Animals were tested individually in two forced swim tests (FSTs) separated by 24 h. The tests were conducted in a Plexiglas cylinder of 45-cm height and 30-cm diameter filled with fresh water (25°) to a height of 15 cm. In the first test, each animal was immersed in water for 15 min, followed 24 h later by a 5-min swim test. After each swim test, the animal was towel dried and placed under a lamp for 30 min for drying. For the swim test, total duration of immobility in the second test was measured. Immobility was defined as floating or remaining motionless without leaning against the wall of the cylinder.

2.5. Open field

A square wooden box (80×80×30 cm) was marked into 64 smaller squares. At the beginning of the 5-min test, each subject was placed on the same corner square and the number of squares crossed and the total duration of immobility were recorded for the next 5 min.

2.6. Elevated plus maze

The wooden maze, elevated from the floor by a 50-cm-tall pedestal, consisted of two open and two closed arms

(each 50 cm long and 10 cm wide) joined together by a central 10×10-cm square block. The closed arms were surrounded by walls of 30-cm height. At the beginning of the testing, each subject was placed at the center of the maze where open and closed arms intersect facing an open arm. The test lasted 5 min and the time spent in the closed arms was recorded.

The open field and elevated plus maze were wiped with an alcohol solution and dried after each use.

2.7. Data analysis

A two-factor (treatment×delay) analysis of variance (ANOVA) was used to compare the behavior of the groups in the three tests described above. A p level of 0.05 or less was employed for significance.

3. Results

3.1. Swim test

Fig. 1 shows duration of immobility (mean±S.E.) in the second swim test for the light-treated and control groups in the two delay conditions. There was a significant effect of treatment [F (1, 28)=13.7, p<0.001] but not of delay on duration of immobility. There was also no significant treatment×delay interaction. The light-treated groups displayed significantly shorter duration of immobility in the second swim test than the controls.

3.2. Elevated plus-maze and open field tests

Table 1 shows the performance of the groups (mean±S.E.) in the elevated plus-maze (time spent in the closed arms) and open field tests (total duration of immobility and the number of squares crossed). In both tests, there was no significant effect of treatment or delay on any of the behavioral measurements nor a significant treatment×delay interaction.

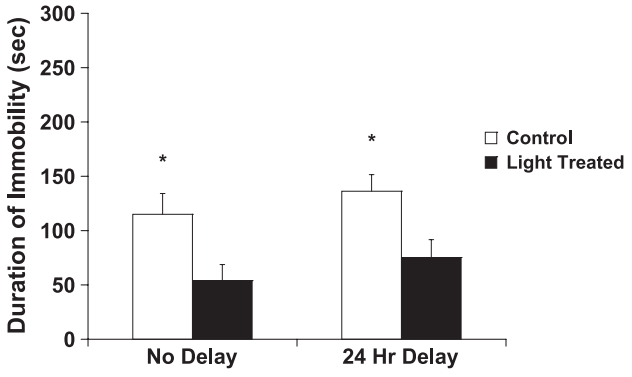


Fig. 1. Duration of immobility (mean±S.E.) in the second of two swim tests separated by 24 h for animals that were either exposed to 12 h light during the dark phase of the 12 h L/12 h D cycle (Light group) or not (Control Group) and tested in two forced swim tests (FSTs) separated by 24 h starting either the same day (No Delay) or 24 h after termination of light treatment (Delay). Asterisks indicate significant difference between light-treated groups and the appropriate controls.

Table 1
Performance (mean±S.E.) of control and light exposed groups that were tested in the open field and elevated plus-maze tests either on the same day light treatment was terminated (No Delay condition) or 24 h thereafter (Delay condition)

		Control	Light treatment
Open field (Duration of immobility)	No Delay	46.8±12.2	33.0±8.6
	Delay	45.8±16.8	41.9±6.2
Open field (Squares crossed)	No Delay	92.5±16.2	64.4±14.7
	Delay	73.0±22.4	73.6±14.7
Plus maze (Closed arm)	No Delay	225.2±41.2	232.5±24.5
	Delay	262.8±14.2	213.1±17.8

4. Discussion

As was hypothesized, the present study showed that rats subjected to a single treatment of light stimulation that converted an 12 h L/12 h D cycle into a 24-h day were protected from behavioral despair as shown by significantly shorter duration of immobility in the second swim test for the light-treated groups compared to the controls. Moreover, the antidepressant effect of such light exposure was still observable 1 day after the light treatment ended during which time animals were maintained in their home cages under the standard 12:12 h L/D lighting schedule. This indicates that the ameliorative effect of a single 12-h light exposure in the dark phase of the L/D cycle light was not due to transient agitation or hyperactivity. This is further supported by the results of the second experiment that showed no difference in the open field and elevated plus-maze behaviors of the light exposed and control subjects in either of the delay conditions. It should be noted that the latter behavioral tests were conducted on days that corresponded to the first and the second swim tests in the first experiment on behavioral despair. On either of the two test days, there was no differential effect of light treatment on the duration of immobility or the number of squares crossed in the open field test. Similarly, light treatment did not alter the time spent in the closed arm of the elevated plus maze, suggesting that such treatment did not have an anxiolytic effect (Pellow et al., 1985). While ethologically more sensitive measurements in the elevated plus maze may reveal an anxiolytic effect of light treatment on anxiety measured in the elevated plus maze (Cole and Rodgers, 1995; Rodgers and Johnson, 1995), the combined results of the open field and elevated plus-maze tests point to a more effective (and positive) effect of light treatment administered in the present study on behavioral despair than on exploratory behavior or anxiety.

The present study suggests an antidepressant-like effect of a single pretreatment with light exposure in behavioral despair of female Wistar rats. Reduced immobility has been repeatedly shown after pretreatment with various types of antidepressants in rats (Armario et al., 1988; Connor et al., 2000; Dalvi and Lucki, 1999; Lucki, 1997; Porsolt et al., 1977, 1978; Reneric and Lucki, 1998). Moreover, Molina-Hernandez and Tellez-Alcantara (2000) have shown that a prolonged administration of rats to a long- (14 h L/10 h D) but not to a short-day (5 h L/19 h D) has protective effect on forced swimming test compared to controls. The ameliorative effect of 30 days of exposure to the long day was comparable to that of daily treatment with clomipramine or desipramine for 30 days. Our results indicate that a single prolonged exposure to light is adequate for an ameliorative effect on behavioral despair as well.

A potential major target for the antidepressant effect of light is the suprachiasmatic nucleus (SCN). SCN is the master pacemaker of the brain that regulates a large number of circadian rhythms (van Esseveldt et al., 2000; Turek, 1985, 1994). Recent studies have begun to discover a role for the SCN in affective functions and disorders (Buijs et al., 1997, 1998, 1999; Engelmann et al., 1998; Van Cauter and Turek, 1986). Studies have uncovered a reciprocal relationship between SCN function and stress-related behaviors in that while stress and psychopathology can impair biological rhythms (Kripke et al., 1978; Meerlo et al., 1996; Stewart et al., 1990a; Tornatzky and Miczek, 1993), disruption of biological rhythms can induce stress (Souetre et al., 1989; Stewart et al., 1990b). More recently, fear has been shown to interfere with light-induced phase changes in SCN functions indicating that SCN is affected by emotional states (Amir and Stewart, 1998).

Neuroanatomical studies have revealed extensive and usually reciprocal relationship between the SCN and several limbic structures that figure critically in affective behavior and psychopathology (Buijs et al., 1998; Leak and Moore, 2001; Vrang et al., 1995; Watts and Swanson, 1987; Watts et al., 1987). The neurochemical basis for the involvement of the SCN in affective behavior and possibly in psychopathology is likely to be related to its mediatory role in the regulation of the hypothalamo-pituitary-adrenal (HPA) axis response to stress (Buijs et al., 1993, 1998, 1999; Kalsbeek et al., 1992; Sage et al., 2001). Moreover, we have recently shown that bilateral destruction of the SCN in male Wistar rats results in reduced immobility in the forced swim tests (Tataroglu et al., 2004).

5. Conclusion

The present findings constitute the first demonstration that a single exposure to light treatment in the dark phase of an L/D cycle can protect rats against the induction of behavioral despair by means of forced swim tests. Our findings may provide the basis for investigating the ameliorative effect of light on depression in an animal model in which the neuroanatomical and neurochemical mechanisms of light action on the appropriate brain structures can be investigated.

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